

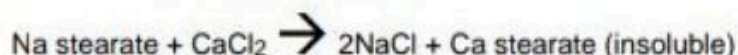
Aim: Determination of total hardness of Water by EDTA (Complexometric) Titration

Purpose: To determine the "hardness" of a sample of water by using EDTA to determine the concentration of calcium and magnesium ions.

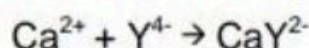
APPARATUS Electronic weighing balance, 250 ml conical flask, Measuring cylinder, funnel, glass rods, Pipettes, Burettes and 100ml beaker

Background

Water hardness refers to the presence of calcium and magnesium salts in water, which react with soap to form an insoluble precipitate:



Water hardness affects the effectiveness of your soap and is responsible for the "scum" in your bathtub if you have hard water. An historical method for determining water hardness was titration with a standard soap solution. We will use a more modern titrant, ethylenediaminetetraacetic acid (EDTA). EDTA is a hexadentate metal ion ligand: it has 6 potential sites for binding a metal ion (4 carboxyl groups and 2 amino groups). Describing EDTA as a chelating agent similarly denotes that it can simultaneously bind a metal ion through more than one binding site. To determine the water hardness, you will determine the total content of Ca^{2+} and Mg^{2+} in the water sample by titrating to form complexes between these metal ions and EDTA. Deprotonated EDTA, given the symbol Y^{4-} , reacts with these ions in the following way:



and



Note that this is 1:1 stoichiometry—the EDTA may be hexadentate, but it is only one EDTA molecule per complex.

You will fix the solution pH at 10 with a pH 10 buffer to ensure that the EDTA is deprotonated. The reaction endpoint will be established using an Eriochrome Black T indicator.

After first determining the total water hardness (assuming that the two target metal ions are the only ions in the sample with which EDTA complexes), you will separately determine the amounts of each of the two metal cations. You will titrate samples to determine the concentration of Ca^{2+} , only, by masking the interference from Mg^{2+} by precipitating it from solution. That is, you will add excess strong base to precipitate Mg^{2+} as insoluble $\text{Mg}(\text{OH})_2$ and then titrate the remaining Ca^{2+} with EDTA. In your analysis, you will assume that no precipitate formed during the pH10 titration, and that errors such as coprecipitation of Ca^{2+} with Mg^{2+} did not occur.

Procedures:

- Pipette out 10 ml of the sample water into a 100 ml conical flask.
- Add NH_4Cl – NH_3 buffer solution 2-3ml to the water sample.
- Add the EBT indicator to the water sample and the colour of the water will turn to wine red.
- Fill up the burette with the supplied EDTA solution of known strength.
- Titrate the water sample against the EDTA drop by drop until the wine red colour turns to blue. Once the blue colour is developed stop the titration.
- Record the burette reading and repeat it twice.

OBSERVATION TABLE

Sr. No	Volume of water sample (ml)	Initial burette reading (ml)	Final burette reading (ml)	Titre reading (Initial – Final) (ml)	Average (ml)
1.	10				
2.	10				
3.	10				

Calculation and Results:

1000 ml 1(M) EDTA $\equiv$ 100gm of CaCO_3

1000ml 1(M) EDTA $\equiv$ 100 x 1000 mg of CaCO_3

1 ml (M/100) EDTA $\equiv$ 1 mg of CaCO_3

X ml (M/100) EDTA $\equiv$ X mg of CaCO_3

10 ml of water sample contains X mg of CaCO_3 1000 ml of water sample contains 100X mg of CaCO_3

Conclusion:

The hardness of the given sample water is 100 X ppm (ppm = mg / litre)

Practice Questions:

1. What is meant by Complexometric Titration?
2. Draw structure of EDTA.
3. Draw Metal-EDTA (1:1) Complex, where $M=Ca^{2+}$, $EDTA^{4-}$
4. What salts are responsible for temporary and permanent hardness of water?
5. Name the method used to remove temporary hardness from water?
6. Name the methods used for removal permanent hardness from water.
7. Except complexometric titration, how can you estimate total hardness present in water?
8. Write unit of hardness.
9. How many potential donor sites present in EDTA.
10. Full form of EDTA and EBT.
11. What is meant by Buffer solution?
12. Name the buffer solution used in this practical and comment on its role.
13. Comment on stability of Calcium-EDTA complex and Calcium-EBT complex.
14. What is meant by Chelate effect?